

Exam Date & Time: 08-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2024

Machine Learning [CSE-4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between Supervised, Unsupervised and Reinforcement Learning. Provide examples of real-world tasks that fall under each category and discuss how machine learning techniques are applied in these scenarios. (5)
 - A)
 - B) Make use of ensemble methods like Random Forests or Gradient Boosting help in reducing overfitting compared to individual models like decision trees. (5)
- 2) Critically assess the limitations of Principal Component Analysis (PCA) in capturing the underlying structure of high-dimensional data. Discuss alternative dimensionality reduction techniques and their potential advantages over PCA in preserving non-linear relationships and handling outliers. (5)
 - A)
 - B) Investigate the interpretability of SVM models trained with different kernels. How do different kernels affect the interpretability of decision boundaries and feature importance in SVM models? Are there trade-offs between model complexity and interpretability associated with kernel selection. (5)
- 3) Evaluate the main steps of the K-means clustering algorithm, examining processes such as initialization, cluster assignment, and centroid updating. (5)
 - A)
 - B) Critically analyze the concept of the bias-variance tradeoff in machine learning. Provide a detailed explanation of how model complexity influences bias and variance, and discuss strategies to strike an optimal balance between them. (5)
- 4) Critically evaluate the suitability of Q-learning for solving complex decision-making problems with large state and action spaces. Discuss its limitations and propose enhancements or alternative algorithms that can overcome these limitations. (5)
 - A)
 - B) Make the convergence properties of the value iteration algorithm. Under what conditions does value iteration converge, and what factors might influence its convergence rate. (5)

- 5) Evaluate the impact of data imbalance on the performance of classification algorithms such as logistic regression and support vector machines. Discuss techniques for handling imbalanced datasets and their implications for model fairness and interpretability. (5)

A)

- B) Consider the tuple $X = (\text{Weather} = \text{Rainy}, \text{Temperature} = \text{Mild}, \text{Humidity} = \text{High}, \text{Windy} = \text{True})$. Calculate the conditional probabilities required for predicting the class label 'PlayTennis' for both 'Yes' and 'No'.

Weather	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes

(5)

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